Written Examination

Health data part

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Written Examination

- **Course:** Health Data and Reproducible Research
- **Duration:** 95 minutes
- **Total marks:** 40

Instructions to students

- Answer all questions.
- Write your answers in **English**.
- In **Section A**, keep your answers brief. In most cases, **1–3 sentences** are sufficient.
- In **Section B**, provide short but clear explanations. A good answer is typically **4–6 sentences**.
- In **Section C**, write a short essay of approximately **half a page to one page**.
- You may use relevant terminology from the course, but make sure your reasoning is clear.
- Where appropriate, motivate your answers briefly.
- The exam is designed to assess both conceptual understanding and the ability to relate course concepts to practical work with health data.

Section A. Conceptual understanding (12 marks)

Answer briefly.

A1. ICD: historical purpose (2 marks)

What was the original purpose of the International Classification of Diseases (ICD) when it was first introduced in the nineteenth century?

A2. Scientific scholarship (2 marks)

Buckheit and Donoho (in one of the papers) argued that a scientific paper is merely *advertising* of the real scholarship. What did they mean by this?

A3. Reproducible pipelines (2 marks)

What problem does a workflow tool such as `{targets}` solve in a data analysis project?

A4. Version control (2 marks)

Explain (at least) one important advantage of using Git and GitHub when developing data analysis scripts.

A5. Quarto (2 marks)

What role does Quarto play in a reproducible research workflow?

A6. Secondary use of health data (2 marks)

What is meant by *secondary use of health data*?

Section B. Applied data management (18 marks)

Provide a short explanation for each question.

B1. Parquet versus CSV (6 marks)

Large health datasets are often distributed as CSV files or SAS data (.sqs7bdat). Explain why this is not ideal and suggest better solutions (motivate why).

B2. ICD revisions (6 marks)

The transition from ICD-9 to ICD-10 greatly increased the number of diagnostic codes. Explain how this affects the *granularity* of clinical research and discuss one potential drawback for epidemiological studies.

B3. Coding inconsistency (6 marks)

Imagine a multicenter observational study based on hospital registry data. Explain how differences in coding practices between hospitals could influence the validity of the study results. (Note that "coding" refers to "medical coding" of medications/conditions/etc; not programming/software development.)

Section C. Critical synthesis (10 marks)

Write a short essay.

C1. The evolving workflow of biostatistical research (10 marks)

Traditionally, a scientific study was often viewed as a final publication. Today, research is better understood as a **dynamic workflow** involving raw registry data, processing pipelines, documentation, and legal constraints.

Discuss how the following elements interact in modern health data research:

- legal frameworks (GDPR)
- data quality and registry data
- medical coding systems such as ICD
- reproducible workflows and computational pipelines

In your answer, address the following question:

How can a researcher balance the need for transparency and reproducibility with practical limitations such as imperfect data quality and legal requirements for data privacy?

Teacher part

The material below is intended for the examiner only and should be removed before distributing the exam to students.

Mark distribution

Section	Maximum marks
A	12
B	18
C	10
Total	**40**

Grade boundaries

Grade	Requirement
Pass (G)	At least**22/40**
Pass with distinction (VG)	At least**32/40 and** at least **6/10 in Section C**

Marking guide

Section A (2 marks each)

A1. ICD: historical purpose
Full marks (2):

- identifies the original purpose as classification of causes of death / mortality statistics
- mentions standardisation or comparability across places or populations

Partial marks (1):

- mentions disease classification in general but not mortality statistics or comparability clearly

No marks (0):

- incorrect or irrelevant answer

A2. Scientific scholarship
Full marks (2):

- explains that the paper is only a summary or advertisement of the work

- identifies code, data, and computational workflow as the real scholarship

Partial marks (1):

- captures only one of the two main points

A3. Reproducible pipelines
Full marks (2):

- mentions automation of workflows, dependency tracking, or avoiding unnecessary reruns
- links this to reproducibility or efficiency

Partial marks (1):

- vague but relevant description of organising analysis steps

A4. Version control
Full marks (2):

- explains one clear benefit such as tracking changes, collaboration, rollback, transparency, or reproducibility

Partial marks (1):

- benefit mentioned but not clearly explained

A5. Quarto
Full marks (2):

- explains that Quarto combines code, text, and results in one document
- links this to reproducible reporting or communication

Partial marks (1):

- mentions reporting or documentation but not reproducibility clearly

A6. Secondary use of health data
Full marks (2):

- explains that data originally collected for one purpose are reused for another purpose
- relevant examples include research, quality improvement, planning, or surveillance

Partial marks (1):

- partially correct but vague

Section B (6 marks each)

B1. Parquet versus CSV
Expected elements:

- Parquet is column-oriented

- only relevant columns need to be read
- more efficient compression
- reduced input/output
- better suited for large-scale querying / analytics

Suggested marking:

- **6:** clear explanation including columnar storage and at least two additional relevant points
- **4:** correct main idea but limited detail
- **2:** fragmented or partially correct
- **0:** incorrect

B2. ICD revisions

Expected elements:

- ICD-10 has many more codes than ICD-9
- greater specificity / finer granularity
- may improve precision in identifying diagnoses or subgroups
- possible drawback: reduced comparability over time, mapping difficulties, changes in coding practice, risk of artefacts in trend analyses

Suggested marking:

- **6:** clear discussion of both increased granularity and at least one substantial drawback
- **4:** correct but underdeveloped
- **2:** only one side addressed
- **0:** incorrect

B3. Coding inconsistency

Expected elements:

- hospitals may code similar cases differently
- can introduce systematic measurement error or misclassification
- can bias comparisons between centres
- may threaten internal validity and interpretation
- can create apparent differences that are artefacts of coding rather than real clinical differences

Suggested marking:

- **6:** clear explanation of mechanism and consequences for validity
- **4:** largely correct but limited depth
- **2:** partial recognition of the issue
- **0:** incorrect

Section C (10 marks)

C1. The evolving workflow of biostatistical research

A strong answer should:

- connect legal, technical, and methodological aspects rather than listing them separately
- discuss GDPR or privacy constraints in relation to access, sharing, and transparency
- address data quality problems such as missingness, coding error, or imperfect documentation
- explain the role of coding systems such as ICD in standardisation and comparability
- describe how reproducible workflows, version control, and computational documents support trust and longevity
- recognise that full openness may not be possible with sensitive health data
- propose realistic alternatives such as sharing code, metadata, synthetic data, documentation, or processing logic

Suggested marking rubric:

Marks	Description
9–10	Excellent synthesis; clear, well-structured, conceptually strong, and addresses the core tension in a nuanced way
7–8	Good analytical answer; connects several themes well, though with less depth or precision
5–6	Reasonable discussion; covers relevant themes but mainly descriptively
3–4	Limited answer; some relevant points but weak structure or shallow reasoning
0–2	Very limited or largely irrelevant

Notes for examiner

- Section A is intended to test breadth and terminology.
- Section B is intended to test applied understanding and the ability to explain mechanisms.
- Section C should carry the main weight for **VG**, since it tests synthesis, judgment, and reflection.
- If desired, minor language errors should not reduce marks as long as the intended meaning is clear.